ARNAV SUMAN

2nd year student @MIT, Bangalore

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Summary

Passionate about leveraging AI/ML to drive positive societal impact. My research interests span across but not limited to diverse domains of AI/ML, including Deep Learning, Computer Vision, Generative Modeling and NLP.

Skills

- Python
- C/C++
- Java
- Tensorflow/Keras
- Data Structures & Algorithums
- Computer Vision
- MySQL
- MERN Stack
- Logical Reasoning

Experience

Ongoing Interdisciplinary Research Sep 2022 – Present

AL/ML Researcher *Bangalore, India*

- Creating a framework to ensure secure transmission and enhancing the security of an Additive Manufacturing network.
- Working on tensorflow library along with creating suitable datasets for the project.
- Trained models to detect errors of < 1 micron in size.
- Working on algorithm to detect triangulation error and code correction.
- Working along my college professor, IIT-Mandi professor and a NIT-Surathkal Master's Student.
- Working towards publishing research paper.

IBM Mentorship and Intel AI Global Impact Festival Finalist Sep 2021 – June 2022

Team Leader *Delhi, India*

- Made an end user Computer Vision Project under the mentorship of IBM mentor to monitor pollution through drones using Deep Learning library.
- Made Desktop App that multiple users can login load a video file or take live input from drone and run Object Detection on top of it and system will automatically detect pollution and then upload that to cloud to be used later all with zero human support.
- Intel AI Impact Creator, Modified Project was selected among top 10 projects from India under Advanced Programming.
- Developed an NLP based chatbot to diagnose possible problems user may face while using our Software.
- Project showcased by Intel on their website and official YouTube of Intel AI Global Impact Festival – India

Raspberry pi Foundation's Competition Coolest Projects.org June 2021

India Winner and Judge's Favorite under Advanced Programming Category *Delhi, India*

- Made chatbots for prevention of cyberbullying, Stuttering. Won Medal

School IT Club President and Hackathon Team Leader Jun 2020 – Jun 2022

- Managed the annual Tech Fest. Headed a team of 21 students from our school, conducted 25 Events. 300+ Students participated from all India.
- School Hackathon Team Leader participated in 15 + Hackathons every year, won many of them.
- Developed an NLP based chatbot to diagnose possible problems user may face while using our Software.

EDUCATION

- **Manipal Institute of Technology, Bengaluru Campus** — Bangalore, India
  B.tech Information Technology - CGPA-8.84 — *Sep 2022 - May 2026*
- **Vishwa Bharati Public School, NCR** — Delhi, India
  CBSE 12th Grade ; Percentage- 85.0% — *April 2014 - July 2022*
  CBSE 10th Grade ; Percentage- 93.2%

PROJECTS

- **Advanced Drone Pollution Monitoring System** – Detect real-time Pollution through Drones using Machine Learning and Computer Vision. (code on GitHub)

- **Climate Change Home Tracker** – Web Based App that informs user of their carbon emissions and footprint predicting carbon emissions in next 50 years and corresponding rise in water level based on current environmental policies. Includes option to change Policy level and visualize change in global earth Temperature. Written in Python using Streamlit library to host on web.

COURSES

- Object-Oriented Hierarchies in Java Object-Oriented Hierarchies in Java (Coursera-LearnQuest)
- Introduction to Object-Oriented Programming with Java (Coursera-LearnQuest)
- Digital Systems: From Logic Gates to Processors (Coursera- Universitat Autònoma de Barcelona)

Profile

- GitHub - https://github.com/arnavsuman
- YouTube - https://www.youtube.com/@arnavsuman3356/featured
- LinkedIn - https://www.linkedin.com/in/arnav-suman/
- Climate Change Project- https://arnavcool123-climate-change-home-tracker-myapp-aysyif.streamlit.app/